

# PHONG TRAN

Full-Stack Software Engineer · TypeScript / Node.js · Microservices · Kubernetes

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## SUMMARY

Full-stack software engineer with 5+ years of professional experience building and running production TypeScript systems at FPT Information System. I own features from technical design to production on a multi-tenant recruitment SaaS of 30+ NestJS microservices deployed for nine enterprise clients, including Shinhan, SHB, Vietbank, HSC and Decathlon: event-driven workflows on NATS and RabbitMQ, MongoDB-to-PostgreSQL data sync, third-party integrations with idempotent retries, and Kubernetes operations. Early adopter of AI-assisted development (Claude Code, Cursor, local LLMs): I plan before I generate, review every line I ship, and stay with a feature until it is verified in production.

## TECHNICAL SKILLS

|                       |                                                                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages             | TypeScript, JavaScript (Node.js), SQL, Python (Flask, pytest), Go                                                                                            |
| Backend               | NestJS, Express, REST API design, Swagger/OpenAPI, JWT/OIDC (Keycloak), multi-tenant SaaS architecture                                                       |
| Messaging & data flow | NATS (request-reply, pub/sub), RabbitMQ, webhooks, cron/batch jobs, ETL, idempotent upserts, retries with backoff                                            |
| Databases             | PostgreSQL (Prisma schemas & migrations), MongoDB (aggregation pipelines), Redis, InfluxDB, MinIO / Azure Blob                                               |
| Cloud & operations    | Kubernetes (kubect!, Helm, rollouts, ConfigMap/Secret), Docker, Docker Swarm, GitHub Actions, OpenTelemetry, Nginx, Linux VPS                                |
| Frontend              | Angular (v12–v21), RxJS, Signals, React, Next.js, Vite, Tailwind CSS, ECharts                                                                                |
| Engineering practice  | Nx / pnpm monorepos, Git with merge-request review, Conventional Commits, Playwright, Agile/Scrum, AI-assisted development (Claude Code, Cursor, local LLMs) |

## PROFESSIONAL EXPERIENCE

Software Engineer – Full-Stack & System Operations Apr 2021 – Present

FPT Information System (FPT IS), Ho Chi Minh City

Product engineer on enterprise HR and smart-traffic platforms; responsible for design, delivery and production operation of my features.

FPT eHiring – multi-tenant recruitment SaaS (ATS) 2022 – Present

NestJS · Angular · Nx · NATS · RabbitMQ · MongoDB · PostgreSQL · Keycloak · Kubernetes · Helm · OpenTelemetry

- Build and operate a recruitment platform of 30+ NestJS microservices and Angular micro-frontends in an Nx monorepo, deployed per tenant for nine enterprise clients across banking, securities and retail: Shinhan Bank, Shinhan Life, Shinhan Finance, SHB, Vietbank, HSC, Y4A, Decathlon and Tan Long Group.
- Designed event-driven workflows on NATS (request-reply and pub/sub) and RabbitMQ for email, CV parsing, document export and integrations; split API services from background workers so long-running jobs scale and fail independently.
- Owned the end-to-end data flow behind the HR analytics dashboard: MongoDB aggregation, batched incremental sync into PostgreSQL with idempotent upserts (INSERT ... ON CONFLICT DO UPDATE), per-record retry and skip-on-failure so one bad record never blocks a run.
- Built the VOffice e-signature integration: adapter pattern for production and mock clients, webhook-driven offer status updates, transient-vs-permanent error mapping, and scheduled monitoring that rehydrates in-flight jobs from the database after pod restarts.
- Implemented tenant isolation across services: Keycloak OIDC/JWT validation via JWKS, tenant resolution by hostname and x-tenant header, per-tenant MongoDB databases, guards, request sanitization and audit interceptors.
- Built a configurable integration engine (Handlebars + JSONata) that maps recruitment events to external HR systems and webhooks, so new client integrations need configuration rather than new code.

- Deploy and run services on Kubernetes (Helm) and Docker Swarm across staging and production per client: build images, manage ConfigMaps/Secrets and rollouts, and debug live incidents from pod logs, events, OpenTelemetry traces and health checks.
- Brought AI-assisted development into the team's workflow as the tools matured: Claude Code and Cursor for specs, scaffolding, tests and log triage, and local LLMs for code that must stay inside client environments. Every generated change is reviewed against the spec for invented fields, wrong data assumptions and missing tenant checks before merge.

### HR Analytics Dashboard

2023 – Present

Angular · ECharts · NestJS · PostgreSQL · MongoDB

- Built NestJS aggregation APIs and an Angular/ECharts dashboard for workforce metrics with filtering and drill-down, aggregating server-side to keep large datasets fast in the browser.

### Violation Detection System (smart traffic)

2021 – 2022

NestJS · Angular · InfluxDB · MongoDB

- Built backend logic that detects rule-based traffic violations from InfluxDB time-series data, plus a dashboard for real-time and historical review.

**Other delivery:** Traffic Estimation System for Binh Dinh (phase 3), real-time device monitoring map, reusable Excel export template service (NestJS), FPT PeopleX. Coordinated a 5-person sub-team on project modules.

### Software Engineer Intern – Back-End

Oct 2019 – Jan 2021

#### FPT Information System (FPT IS), Ho Chi Minh City

- Built REST APIs in Node.js (Express, NestJS) for traffic management and vehicle tracking systems, working with InfluxDB time-series data and MongoDB.

## SELECTED PROJECTS

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### Clinic CRM & Operations Platform (freelance)

Jul 2026 – Present

NestJS · RabbitMQ · PostgreSQL/Prisma · MongoDB · Redis · React · Playwright · Docker · GitHub Actions

- Designed and built an 8-service NestJS backend behind an API gateway (JWT + RBAC) that talks to domain services over RabbitMQ, with a PostgreSQL schema per service and MongoDB for document data, covering CRM, appointments, medical records, payments, invoices and memberships.
- Added an asynchronous audit trail (gateway interceptor → queue → audit service), inbound lead connectors for third-party CRMs (Getfly, Pancake), and an LLM copilot behind an OpenAI-compatible interface with a static fallback; ship via GHCR images and a GitHub Actions deploy pipeline.

### POSM Resize Tool – licensed desktop app (Python)

Sep 2026 – Present

Python · Flask · MongoDB · pytest · GitHub Actions

- Turns monthly Excel order sheets into print-ready PDF, PDF/X-1a and SVG banners from vector templates, replacing manual resizing in Adobe Illustrator; Windows and macOS builds produced in CI.
- Built online license activation with a Flask key-management server, per-machine binding and an admin console; covered by a pytest suite across parsing, export and licensing.

### x-it – open-source developer operations portal

[github.com/fontete/x-it](https://github.com/fontete/x-it) · 2026

Go · React · Vite · Tailwind CSS

- Single Go binary with an embedded React UI that manages Claude Code profiles, skills and plugins, SSH hosts, port forwards, Docker and WireGuard peers; loopback-only API with a DNS-rebinding/CSRF guard, validation and a backup before every config write.

## EDUCATION

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**Bachelor of Science in Computing** – University of Greenwich (Vietnam campus)

2017 – 2021

## CERTIFICATIONS & LANGUAGES

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Angular – The Complete Guide (Udemy) · The Complete JavaScript Course: From Zero to Hero (Udemy)

English – Professional working proficiency · Vietnamese – Native